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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 05/28/2007 |
| **PI Name:** | Chang, Hsueh-Chia Ph.D. | | |
| **Project Title:** | Microcirculation Anomalies in Microgravity Blood Flow | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FLUID PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Complex fluids | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | UNIVERSITY | **Phone:** | 574 631 5697 |
| **Organization Name:** | Notre Dame University | | |
| **PI Address 1:** | 165A Fitzpatrick Engr. | | |
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| **PI Web Page:** | | | |
| **City:** | Notre Dame | **State:** | IN |
| **Zip Code:** | 46556 | **Congressional District:** | 3 |
| **Comments:** | | | |
| **Project Type:** | INTERDISCIPLINARY (2004-2007) | **Solicitation / Funding Source:** | 2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities |
| **Start Date:** | 02/04/2002 | **End Date:** | 10/31/2007 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 4 |
| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Ansari, Rafat | **Contact Phone:** | 216-433-5008 |
| **Contact Email:** | Rafat.R.Ansari@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received NCE to 10/31/2007 per PI (11/06) | | |
| **Key Personnel Changes/Previous PI:** | 0 | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NAG 3 2701 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | The project is to study anomalies in micro-gravity blood flow in human circulation system and in blood diagnostic kits. We have solved an important blood suspension loading problem into small diagnostic kits and are now working on several chip-scale blood analytical tools. Included are blood coagulation detecion, blood glucose detection, leukemia, sickle cell and blood bacteria detection. The target is to produce simple kits that do not require elaborate equipment that is unavailable in space.<br><br>To develop blood diagnostic kits that can be used in a confined spaceship. The diagnostic kits would be to detect pathogens and anomal physiological symptons duirng prolonged space flights.<br><br>It can produce a robust and miniature bloood diagnostic kit for a variety of blood anomalies and diseases often associated with space flights. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The same blood diagnostic kits could be used as point-of-care and portable diagnostic kits on earth. |
| **Task Progress:** | The NASA funding is used to pursue blood diagnostic kit design. It employs dielectrophoresis (DEP) to sort blood cells of different age. A novel buffer is developed for the blood cells to select optimum dielectric relaxation times for the solution and a gludaraldehyde crosslinking agent is added to change the dielectric permittivity of the blood cell membrane to effect sensitive dielectrophoretic analysis and discrimination of bovine red blood cell starvation age. The age of the blood cell is found to be affect the surface protein density which, in turn, changes its dielectric relaxation time. The buffer is selected to amplify the sensitivity of the DEP cross-over frequency to the membrane protein density. This solution can discriminate blood cells between 1 week and 4 weeks in age. The DEP assay is carried out in a diagnostic kit designed in the PI's group and, with proper optimization, it can determine the age of the blood cell within 10 minutes (1). It is also believed that the same assay can be applied to detect other blood cell diseases in space.<br>To detect blood cell infection by bacteria, the PI's group has also developed an anti-biotic screening method, coupled with an on-chip impedance bacteria quantification technique to identify the bacteria responsible for septic infection of the blood cells (2).<br><br>(1) Cheng, I-F.,Chang, H.-C., Hou, D. and Chang, H.-C., "An Integrated Dielectrophoretic Chip for Continuous Bioparticle Filtering, Focusing, Trapping and Detecting", Biomicrofluidics, 1, 021503(2007)<br><br>(2) Sengupta, S, Battigelli, D. A. and Chang, H.-C., "A micro-scale multi-frequency reactance measurment technique to detect bacterial growth at low-bioparticle concentrations", Lab-on-a-Chip, 6,682(2006) |
| **Bibliography Type:** | Description: (Last Updated: 02/21/2014) |
| **Articles in Peer-reviewed Journals** | Cheng I-F,Chang H-C, Hou D, Chang H-C. "An integrated dielectrophoretic chip for continuous bioparticle filtering, focusing, trapping and detecting." Biomicrofluidics, 2007 Jan;1:021503. In press, January 2007. , Jan-2007 |
| **Articles in Peer-reviewed Journals** | Sengupta S, Battigelli DA, Chang HC. "A micro-scale multi-frequency reactance measurement technique to detect bacterial growth at low bio-particle concentrations." Lab Chip. 2006 May;6(5):682-92. PMID: 16652185 , May-2006 |